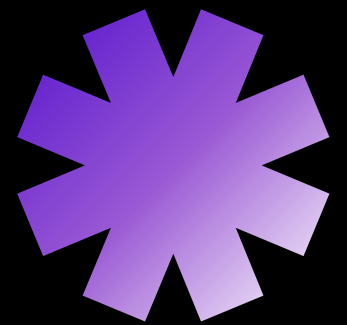
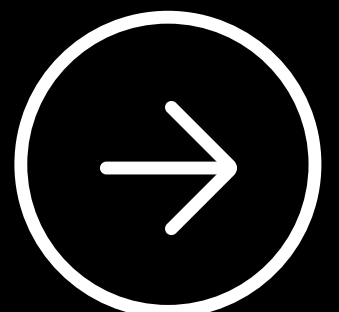




DEVOPS PRACTICE

DAY 11 – DAY 13

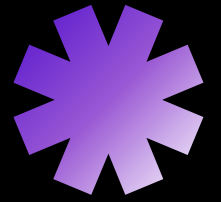
- Linux file ownership and permissions
- User and group management
- Linux service troubleshooting
- LVM storage management
- Hands-on Linux and DevOps practice
- Learning by doing in Linux environment



11

- *ls -l command*
- *chmod permissions*
- *chown ownership*
- *chgrp command*
- *Recursive ownership*
- *User and group management*
- *Linux access control*

```
ubuntu@ip-172-31-1-155:~$ touch bank-heist/access-codes.txt
ubuntu@ip-172-31-1-155:~$ touch bank-heist/blueprints.pdf
ubuntu@ip-172-31-1-155:~$ touch bank-heist/escape-plan.txt
ubuntu@ip-172-31-1-155:~$ sudo chown tokyo:vault-team access-codes.txt
chown: cannot dereference 'access-codes.txt': No such file or directory
ubuntu@ip-172-31-1-155:~$ sudo chown tokyo:vault-team bank-heist/access-codes.txt
ubuntu@ip-172-31-1-155:~$ ls -l bank-heist
total 0
-rw-rw-r-- 1 tokyo vault-team 0 Jun  1 17:26 access-codes.txt
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun  1 17:27 blueprints.pdf
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun  1 17:27 escape-plan.txt
ubuntu@ip-172-31-1-155:~$ sudo chown berlin:tech-team bank-heist/blueprints.pdf
ubuntu@ip-172-31-1-155:~$ ls -l bank-heist
total 0
-rw-rw-r-- 1 tokyo vault-team 0 Jun  1 17:26 access-codes.txt
-rw-rw-r-- 1 berlin tech-team 0 Jun  1 17:27 blueprints.pdf
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun  1 17:27 escape-plan.txt
ubuntu@ip-172-31-1-155:~$ sudo chown nairobi:vault-team bank-heist/escape-plan.txt
ubuntu@ip-172-31-1-155:~$ ls -l bank-heist
total 0
-rw-rw-r-- 1 tokyo vault-team 0 Jun  1 17:26 access-codes.txt
-rw-rw-r-- 1 berlin tech-team 0 Jun  1 17:27 blueprints.pdf
-rw-rw-r-- 1 nairobi vault-team 0 Jun  1 17:27 escape-plan.txt
```

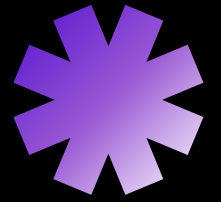


Revision and Linux practice day.

12

- Linux command revision
- File operations practice
- User and group creation
- Service troubleshooting
- systemctl and journalctl
- chmod and chown practice
- Linux troubleshooting confidence

```
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 2 15:09 bank-heist
-rw-rw-r-- 1 berlin ubuntu 0 Jun 1 16:51 devops-file.txt
drwxrwxr-x 4 professor planners 4096 Jun 1 17:12 geist-project
-rw-rw-r-- 1 ubuntu ubuntu 41 Jun 1 15:27 notes.txt
-rw-rw-r-- 1 berlin heist-team 0 Jun 1 17:04 project-config.yaml
-rwxrwxrwx 1 ubuntu ubuntu 20 Jun 1 15:28 script.sh
-rw-rw-r-- 1 deep dev 0 Jun 1 16:59 team-notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Jun 2 15:31 test-note.txt
ubuntu@ip-172-31-1-155:~$ sudo chown noumu:test test-note.txt
ubuntu@ip-172-31-1-155:~$ ls -l
total 16
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 2 15:09 bank-heist
-rw-rw-r-- 1 berlin ubuntu 0 Jun 1 16:51 devops-file.txt
drwxrwxr-x 4 professor planners 4096 Jun 1 17:12 geist-project
-rw-rw-r-- 1 ubuntu ubuntu 41 Jun 1 15:27 notes.txt
-rw-rw-r-- 1 berlin heist-team 0 Jun 1 17:04 project-config.yaml
-rwxrwxrwx 1 ubuntu ubuntu 20 Jun 1 15:28 script.sh
-rw-rw-r-- 1 deep dev 0 Jun 1 16:59 team-notes.txt
-rw-rw-r-- 1 noumu test 0 Jun 2 15:31 test-note.txt
ubuntu@ip-172-31-1-155:~$
```



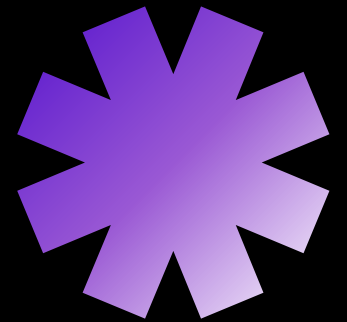
13

Learning Linux Volume Management (LVM).

- Physical Volume (PV)
- Volume Group (VG)
- Logical Volume (LV)
- Mounting storage
- Extending storage dynamically
- Linux storage management
- Hands-on LVM practice

```
root@ip-172-31-1-155:~# df -h /mnt/app-data
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/devops--vg-app--data 179M    152K   168M   1% /mnt/app-data
root@ip-172-31-1-155:~# df -h
Filesystem                Size      Used Avail Use% Mounted on
/dev/root                  6.7G    2.3G    4.4G  35% /
tmpfs                      455M         0   455M   0% /dev/shm
tmpfs                      182M    912K   181M   1% /run
efivarfs                   128K     3.3K   120K   3% /sys/firmware/efi/efivars
tmpfs                      455M    174M   281M  39% /tmp
none                       1.0M         0    1.0M   0% /run/credentials/systemd-journald.service
ld.service
none                       1.0M         0    1.0M   0% /run/credentials/systemd-resolved.service
ed.service
/dev/nvme0n1p13            989M     96M   826M  11% /boot
/dev/nvme0n1p15           105M     6.3M    99M   7% /boot/efi
none                       1.0M         0    1.0M   0% /run/credentials/systemd-networkd.service
kd.service
none                       1.0M         0    1.0M   0% /run/credentials/serial-getty@ttyS0.service
tyS0.service
none                       1.0M         0    1.0M   0% /run/credentials/getty@tty1.service
vice
tmpfs                      91M     8.0K    91M   1% /run/user/1000
/dev/mapper/devops--vg-app--data 179M    152K   168M   1% /mnt/app-data
```


Consistency Daily Practice Hands-On Learning



- Linux permissions and ownership
- User and group management
- Service troubleshooting
- Storage management with LVM
- Real Linux server practice

```
root@ip-172-31-1-155:~# pvcreate /dev/loop3
Physical volume "/dev/loop3" successfully created.
root@ip-172-31-1-155:~# pvs
PV          VG Fmt  Attr PSize   PFree
/dev/loop3  lvm2 ---  300.00m 300.00m
root@ip-172-31-1-155:~# vgcreate devops-vg /dev/loop3
Volume group "devops-vg" successfully created
root@ip-172-31-1-155:~# vgs
VG          #PV #LV #SN Attr   VSize   VFree
devops-vg   1   0   0 wz--n- 296.00m 296.00m
root@ip-172-31-1-155:~# lvcreate -L 150M -n app-data devops-vg
Rounding up size to full physical extent 152.00 MiB
Logical volume "app-data" created.
root@ip-172-31-1-155:~# lvs
LV          VG          Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Con
vert
app-data    devops-vg   -wi-a----- 152.00m
```